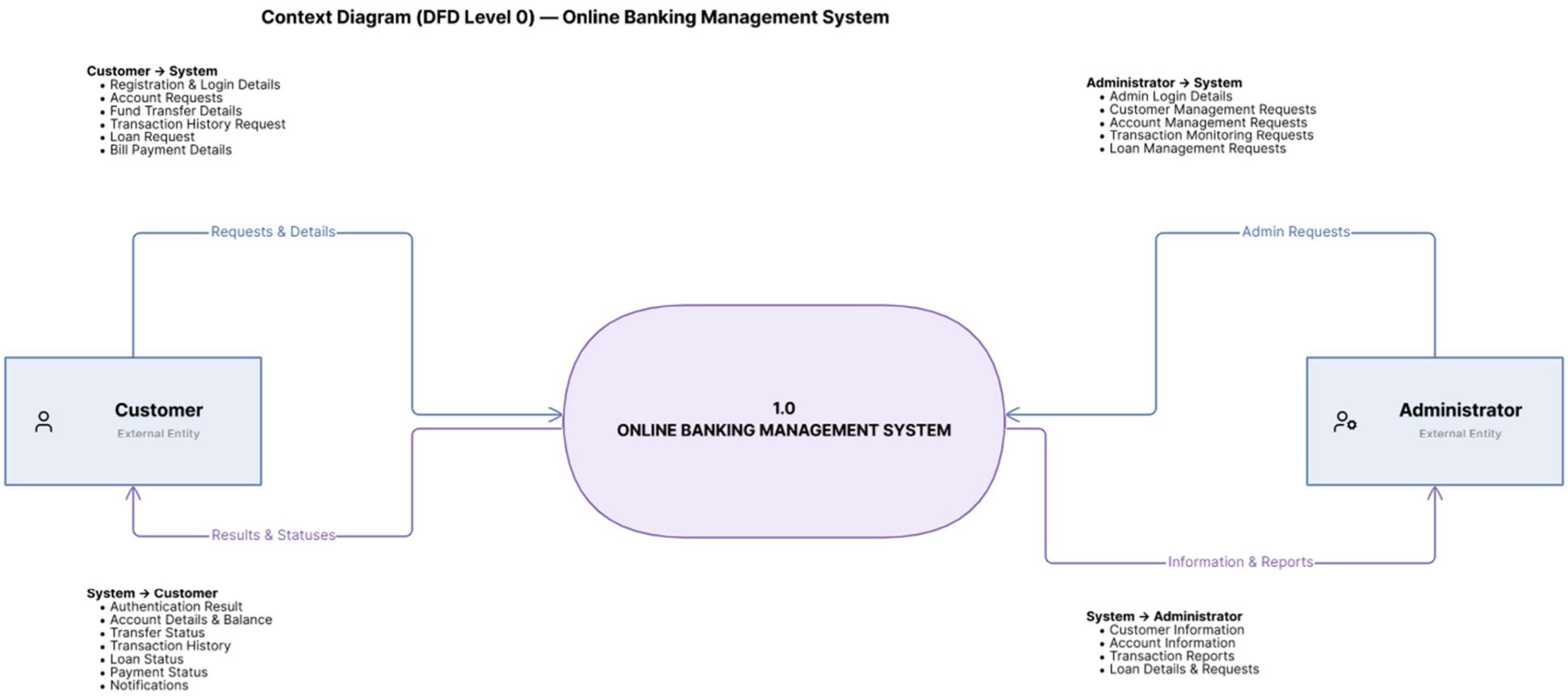
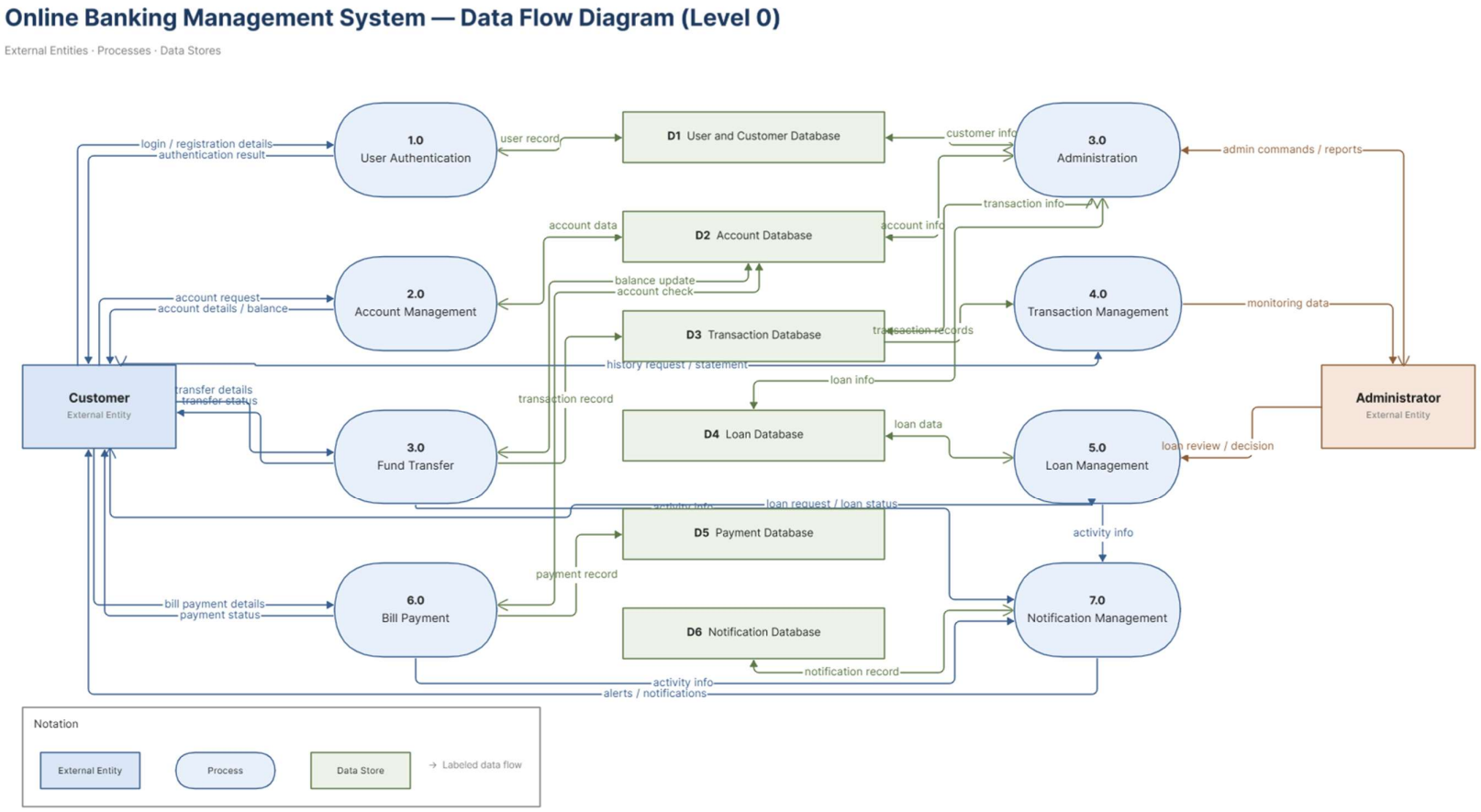
Lab 04 : Design Phase1: High Level Design using Function-Oriented Design Approach

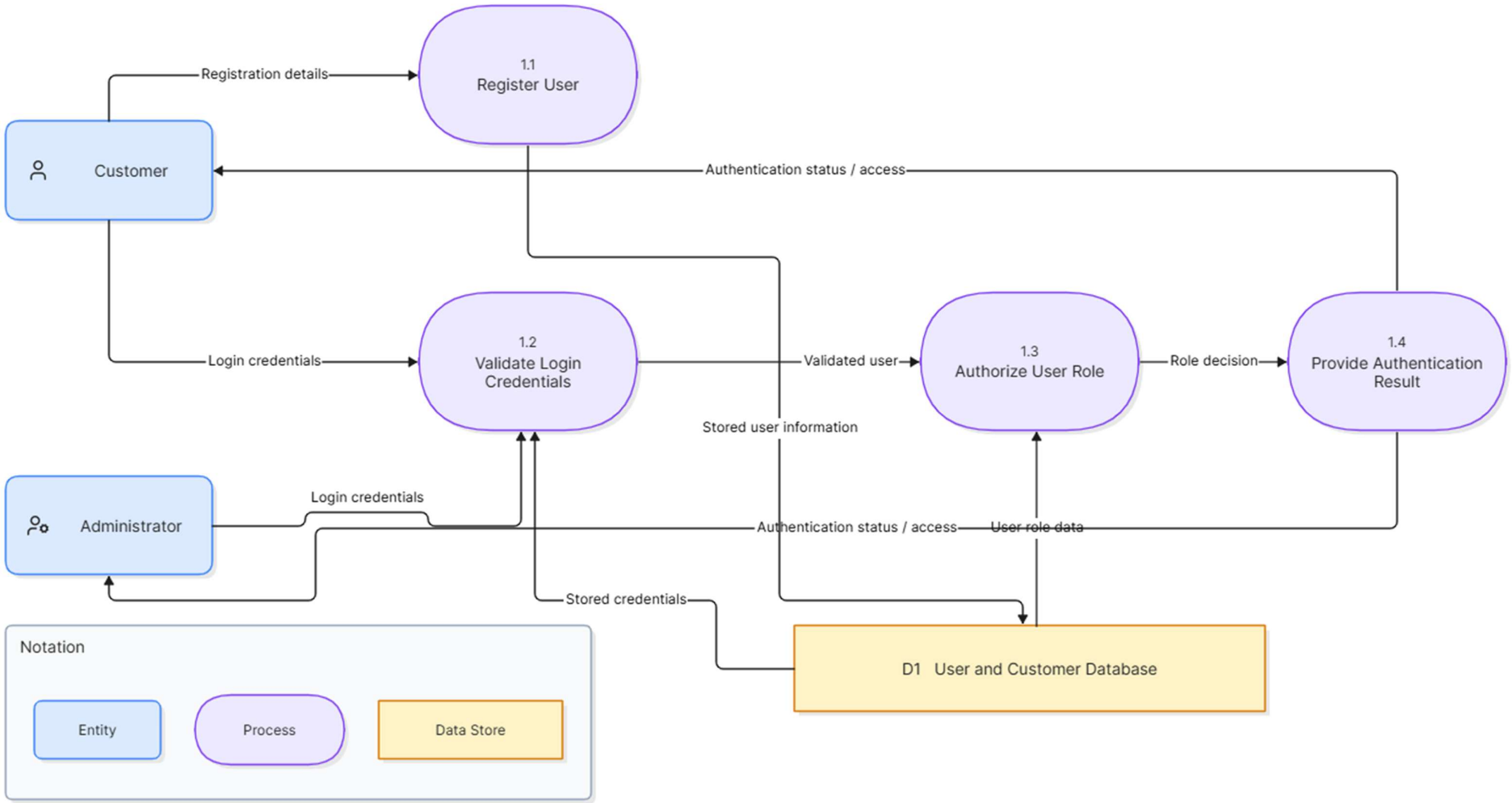
CONTEXT DIAGRAM:



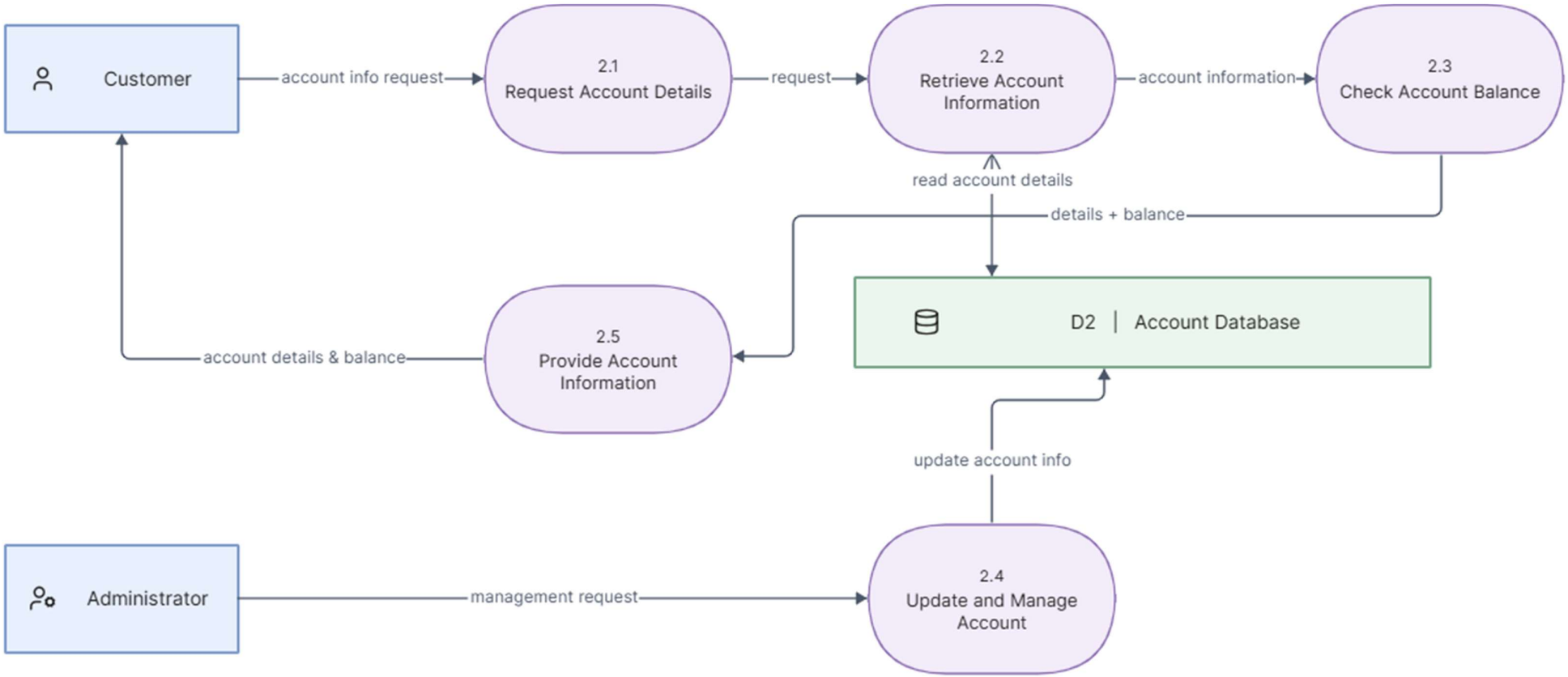
DFD LEVEL 0:


Online Banking Management System — Data Flow Diagram (Level 0)
External Entities · Processes · Data Stores
1.0
User Authentication
2.0
Account Management
3.0
Fund Transfer
4.0
Transaction Management
5.0
Loan Management
6.0
Bill Payment
7.0
Notification Management
3.0
Administration
D1 User and Customer Database
D2 Account Database
D3 Transaction Database
D4 Loan Database
D5 Payment Database
D6 Notification Database
Customer
External Entity
Administrator
External Entity
login / registration details
authentication result
user record
customer info
admin commands / reports
transaction info
account data
account info
account request
account details / balance
balance update
account check
monitoring data
transaction records
history request / statement
loan info
transfer details
transfer status
transaction record
loan data
loan review / decision
loan request / loan status
activity info
payment record
bill payment details
payment status
notification record
activity info
alerts / notifications
Notation
External Entity
Process
Data Store
Labeled data flow

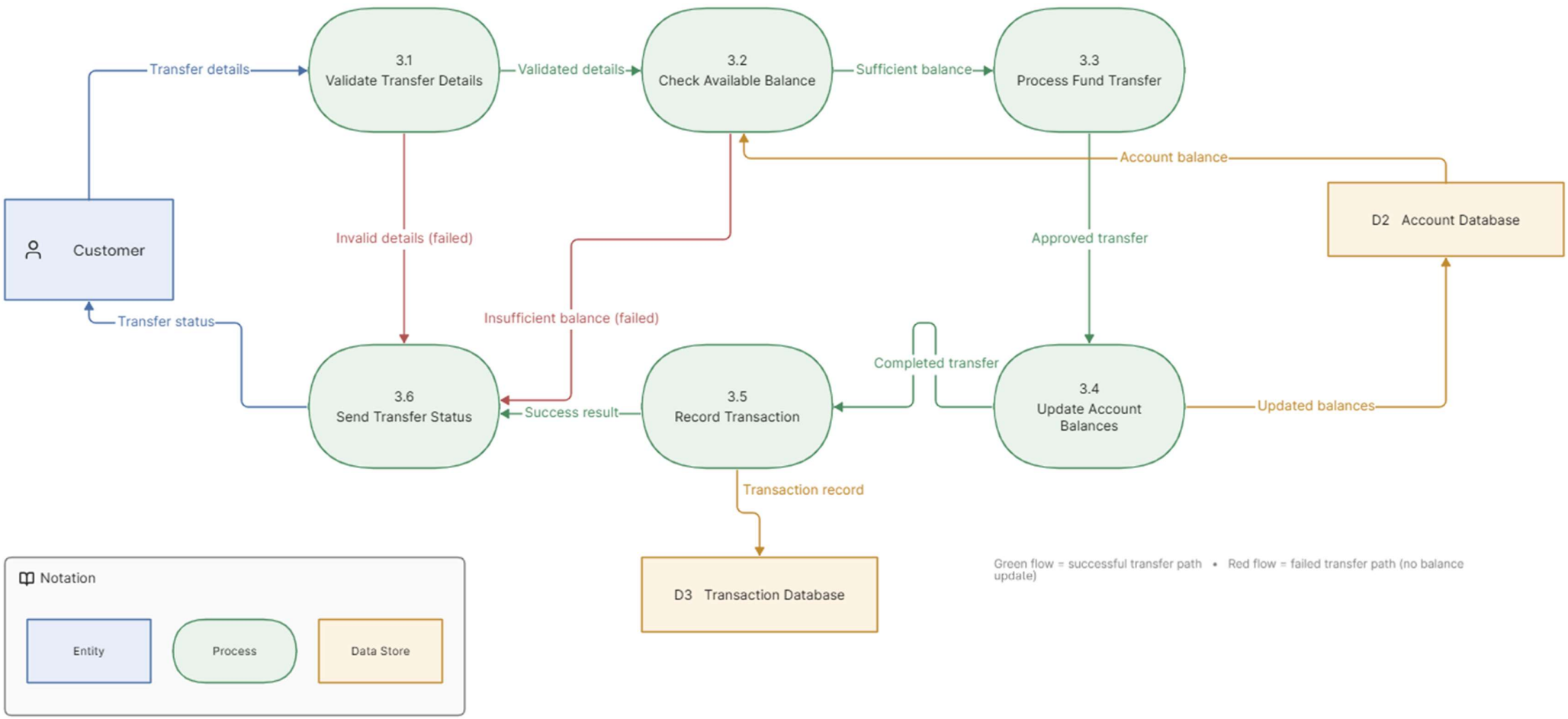
DFD Level 1 — Process 1.0 User Authentication
Online Banking Management System



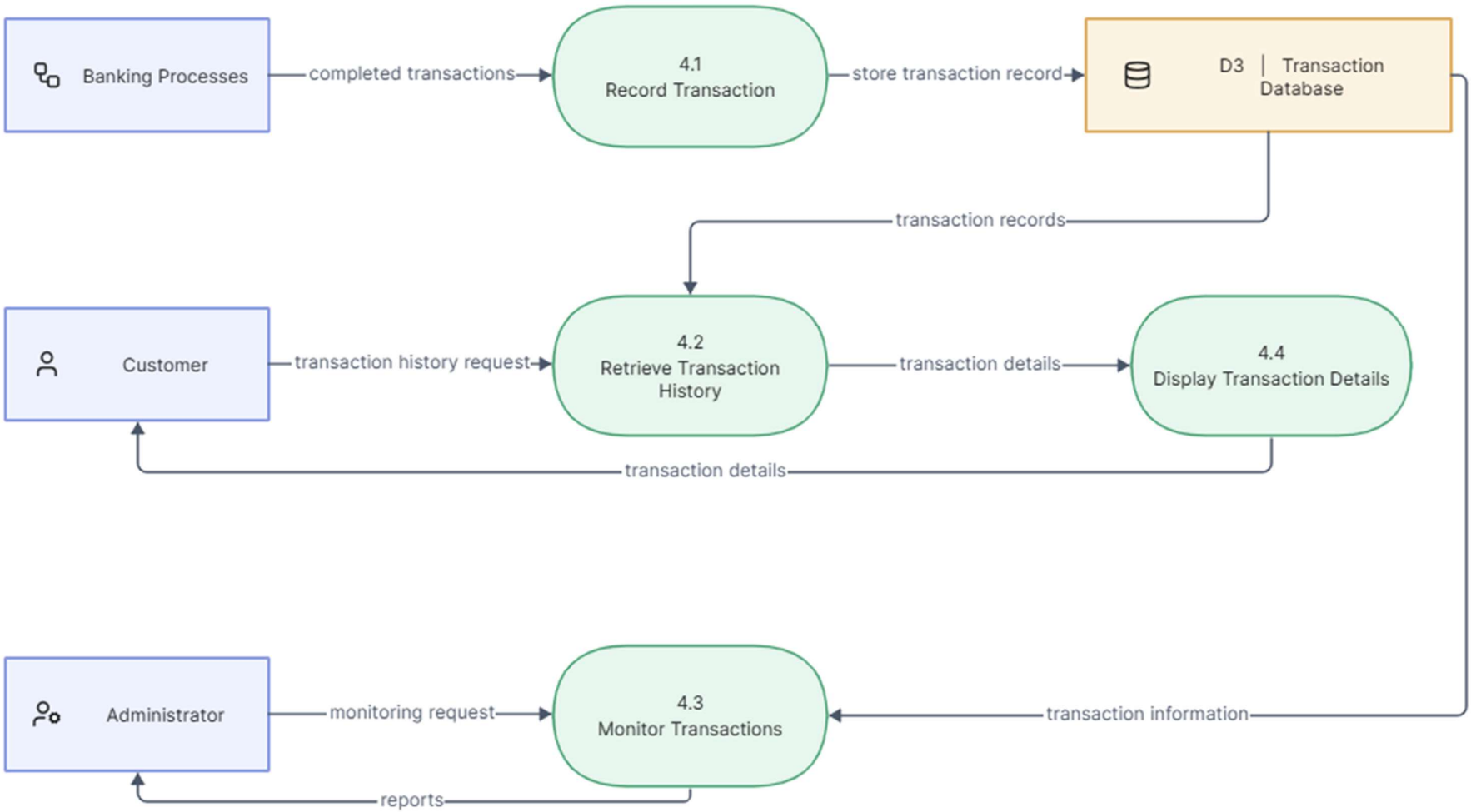
DFD Level 1 — Process 2.0 Account Management
Online Banking Management System



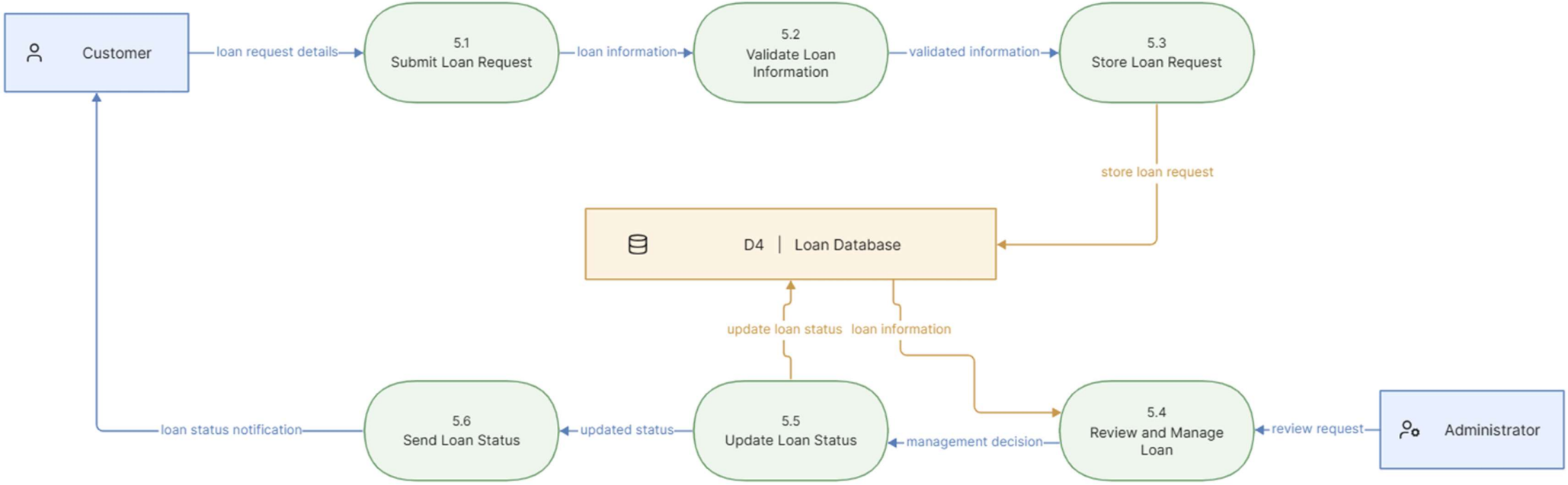
DFD Level 1 — Process 3.0 Fund Transfer
Online Banking Management System



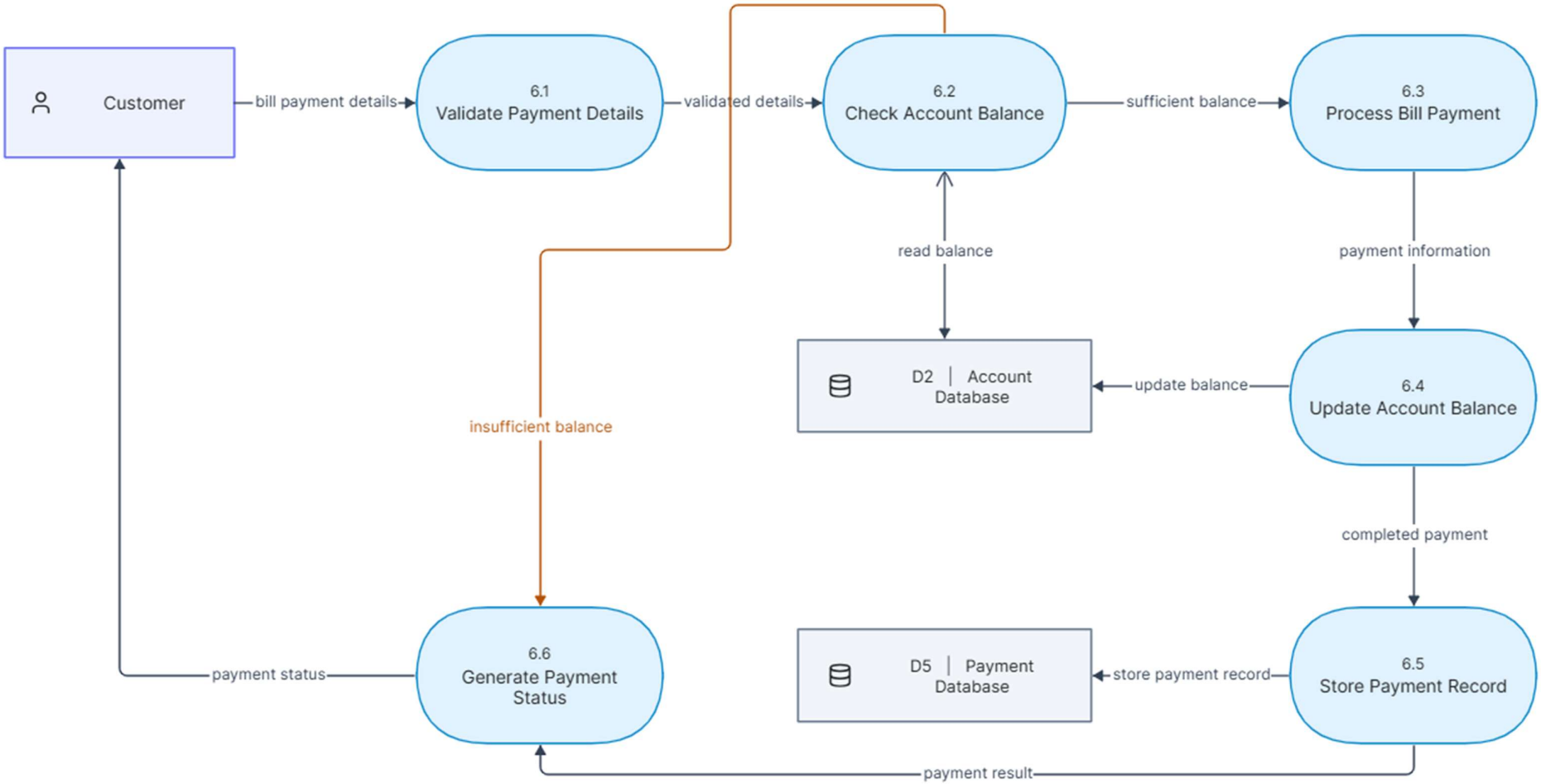
DFD Level 1 — Process 4.0 Transaction Management
Online Banking Management System



DFD Level 1 — Process 5.0 Loan Management
Online Banking Management System

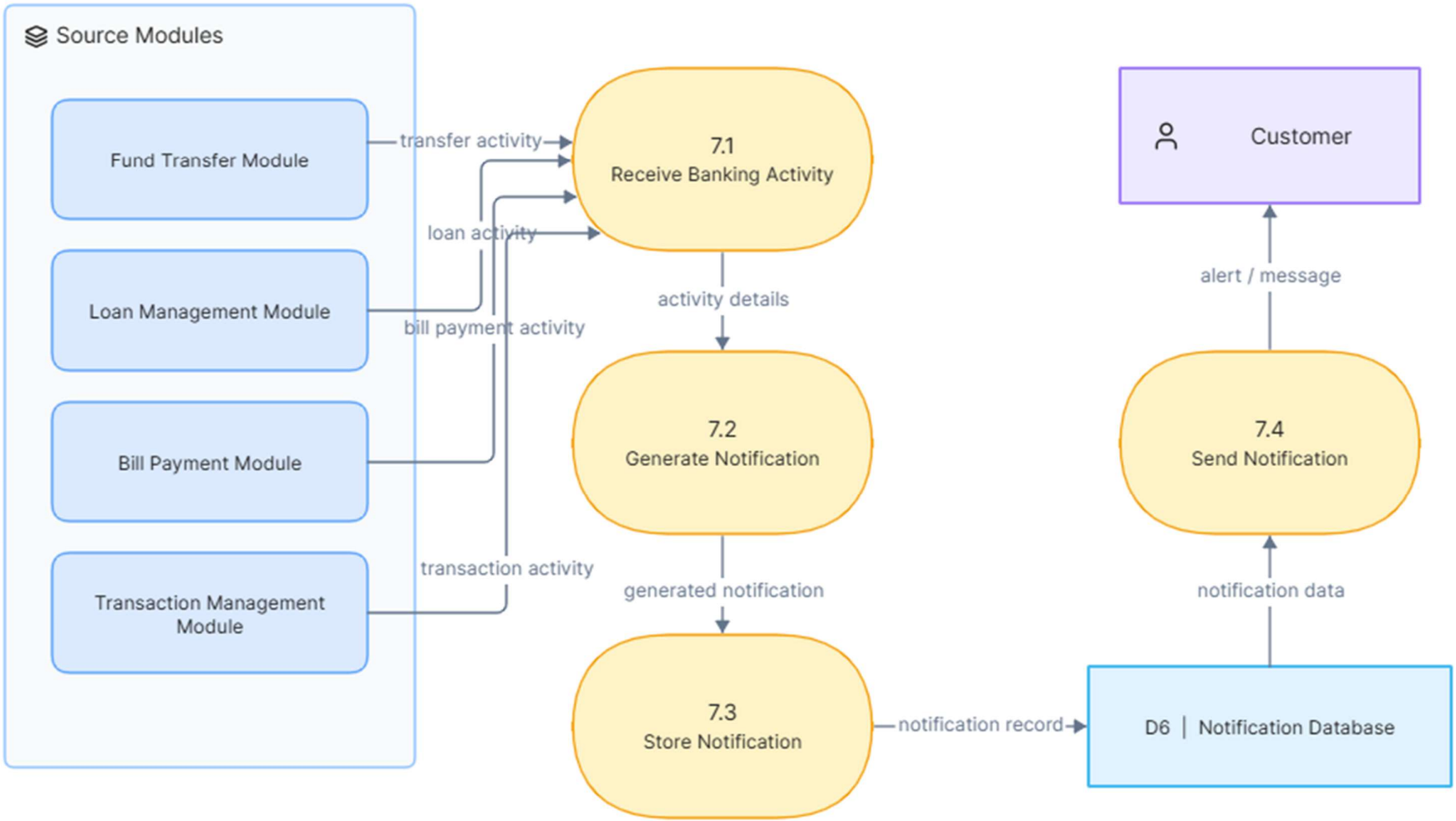


DFD Level 1 — Process 6.0 Bill Payment
Online Banking Management System

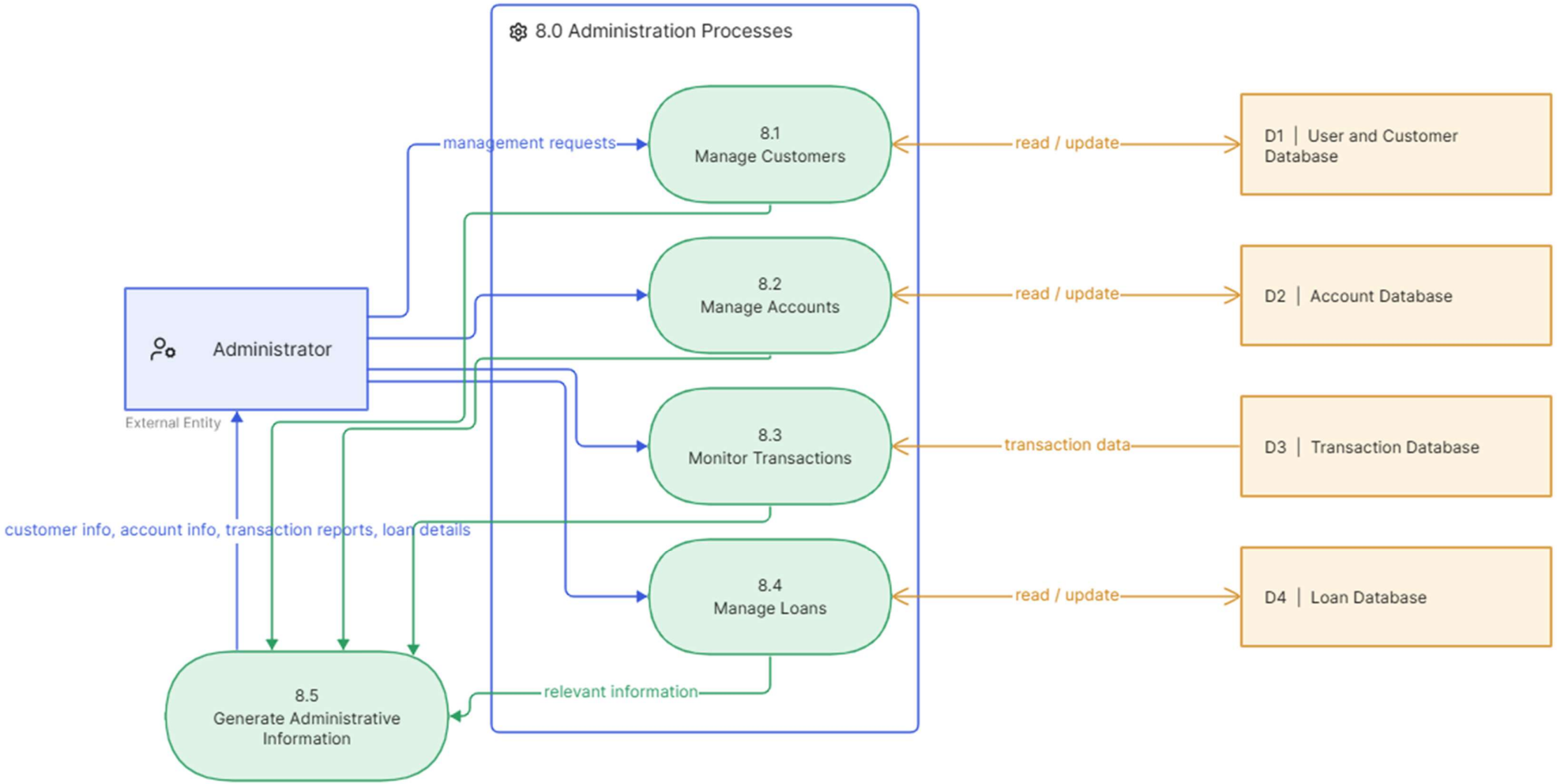


Insufficient balance bypasses 6.3–6.5; no update is made to D2 and a failed payment status is returned.

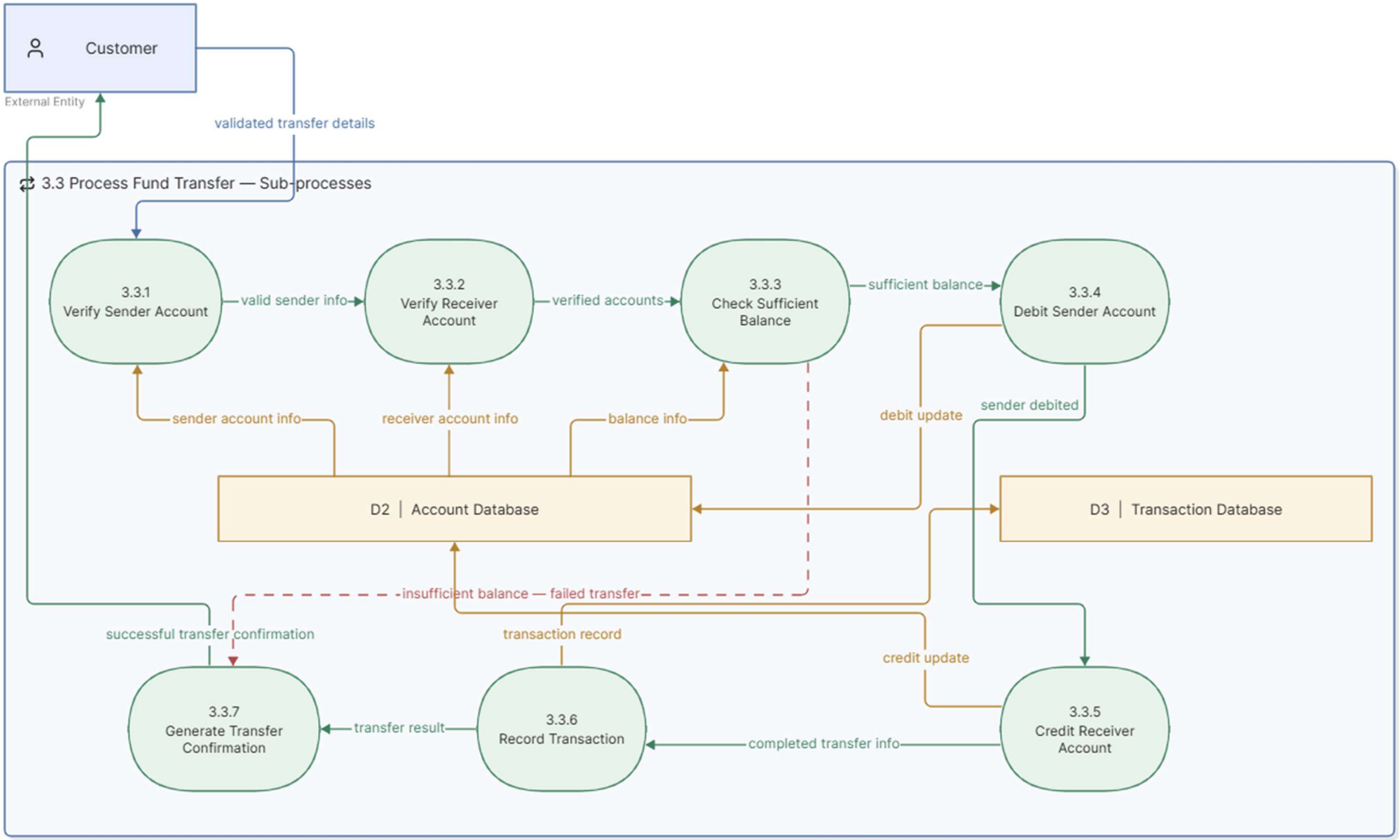
DFD Level 1 — Process 7.0 Notification Management



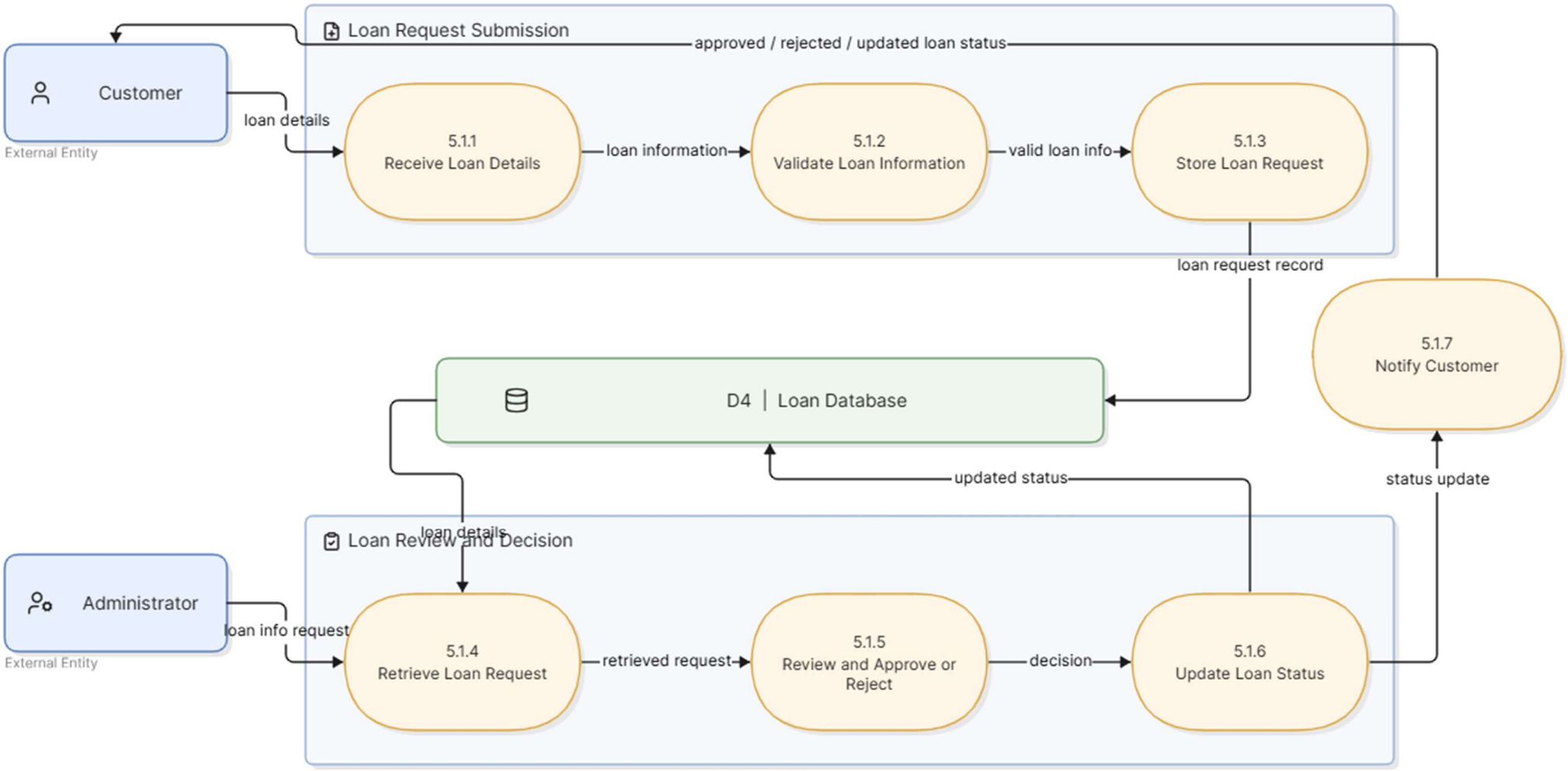
DFD Level 1 — Process 8.0 Administration



DFD Level 2 — Process 3.3 Process Fund Transfer



DFD Level 2 — Process 5.1 Loan Request and Review



DFD Level 2 — Process 6.3 Process Bill Payment

